

Options

ID: aoa_estimation_bladerf
Title: BladeRF ...Root-MUSIC)
Author: DoA Thesis Project
Description: Real-...ot-MUSIC
Output Language: Python
Generate Options: QT GUI

Variable

ID: center_freq
Value: 2.42e9=2.42G

Variable

ID: sample_rate
Value: 1e6=1M

QT GUI Range

ID: rx_gain
Label: RX Gain (dB)
Default Value: 40
Start: 0
Stop: 60
Step: 1

QT GUI Range

ID: phase_cal_deg
Label: Phase Cal (deg)
Default Value: 0
Start: -180
Stop: 180
Step: 100m

Variable

ID: tone_freq
Value: 50e3=50k
TX tone offset frequency

Variable

ID: snapshot_size
Value: 1.024k
Samples per MUSIC snapshot (per paper: 1024)

Variable

ID: antenna_spacing
Value: 500m
Normalized spacing d/lambda = 0.5 for lambda/2

Variable

ID: phase_cal_rad
Value: phase_cal_deg / 180.0=0
Enter the calibration coefficient from phase_calibration flowgraph (2022)\nUses Root-MUSIC algorithm

Complexity: 254ubalAoA estimation flowgraph for BladeRF 2.0 MIMO\nBased on Wachowiak & Kryszkiewicz (2022)\nUses Root-MUSIC algorithm

